

Sujay Aryal

AI ENGINEER · FULL-STACK & MOBILE

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linkedin.com/in/sujayaryal · Open to remote – IST, with EU / US overlap

PROFILE

AI and software engineer with an MSc in Data Science (Merit) and six years building and shipping production software. Sole architect and owner of a hospital ERP running live in Nepal, built on Next.js, TypeScript and PostgreSQL; day-to-day ownership of a multi-tenant feedback platform and a three-app mobile portfolio whose flagship holds 100,000+ downloads across Android and iOS. Master's research on reducing hallucination in Graph RAG retrieval. Works end to end — retrieval design and data modelling through to deployment, releases and the Linux servers underneath.

CORE SKILLS

AI & ML	Large Language Models · Retrieval-Augmented Generation · Graph RAG & subgraph retrieval · prompt design · evaluation of generated output · deep learning (CNNs) · Python data pipelines
Languages	TypeScript · JavaScript · Python · SQL · PHP
Frameworks	Next.js · React.js · React Native · Node.js / Express · Prisma ORM · React Query · Tailwind CSS · Drupal
Infrastructure	PostgreSQL · MySQL · Linux / Ubuntu administration · VPS deployment & process management (PM2) · multi-tenant architecture · Play Store & App Store release pipelines
Practice	End-to-end delivery ownership · release planning & issue tracking (Plane) · developer coordination & bug triage · ticket-based support · client-facing consulting

EXPERIENCE

Software Engineer – ERP & Operational Systems · Nova Hospital, Dhangadhi (freelance, remote) Apr 2026 – Present

- Sole owner of architecture, deployment and ongoing maintenance. Next.js 14 · TypeScript · Express.js · Prisma ORM · PostgreSQL · Tailwind CSS · React Query · PM2.
- Supply chain.** Built the end-to-end goods pipeline — incoming goods across normal, lab and fixed-asset classes, purchase orders, vendor management and bulk Excel/CSV import.
 - Stock control.** Built a real-time stock register covering opening balances, multi-item dispatch, low-stock thresholds and expiry alerting.
 - Approval workflow.** Implemented multi-level requisition and approval with PIN-based storekeeper release, so stock movement is authorised and auditable.
 - Localisation & reporting.** Integrated the Nepali Bikram Sambat calendar, VAT reporting and analytics dashboards to meet local statutory requirements.
 - Operate it.** Deployed on VPS under PM2 and maintain it in production, iterating directly on feedback from hospital staff using it daily.

Software Engineer – Applications & Delivery · Vipassana Research Institute, Mumbai Mar 2020 – Present

PROGRESSION – SIX YEARS, FOUR INTERNAL ROLES

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|----------|--|
| Feb 2025 | App Project Coordinator, Archives & Server Management — app delivery ownership, archives, application server operations |
| Mar 2023 | App Development & Server Management — React Native development, deployments, scheduled-job automation |
| Mar 2021 | Archives, A/V & Web Development — archival digitisation; audio/video production for major events; Drupal |
| Mar 2020 | Volunteer — web and media support |
- Multi-tenant platform.** Designed and shipped a centralised QR-based feedback platform in React.js and Node.js serving five distinct units — pagoda, meditation centre, guest house, souvenir shop and food court — as one configurable system rather than five separate tools, across a site taking 2,000–3,000 visitors daily and up to 20,000 on festival days.
 - Automated reporting.** Replaced manual trustee reporting with automated weekly summary emails and an admin analytics dashboard covering rating trends — current data without anyone compiling it by hand.

- **Application infrastructure.** Own the Linux servers behind the applications — deployments, scheduled jobs and cron automation, configuration and monitoring.
- **Device & server provisioning.** Build, configure and dispatch the course-critical units centres run on — Dëshna (multilingual audio), Gong automation, CCD and course servers, VCM tablets — imaged in-house, then couriered out as ready-to-run installations.
- **Mobile portfolio.** Own three production React Native applications on Play and the App Store — Vipassana Meditation, AT Vipassana and Global Vipassana Pagoda — covering feature development, store releases and live troubleshooting. The flagship carries **100,000+ Play Store downloads, rated 4.7★ across 5,000 reviews on Android and 4.5★ on iOS.**
- **Dipi registration platform.** Administer the course registration system used across centres — onboarding new centres and users, developing the frontend, and resolving issues through a shared ticketing queue.
- **Delivery ownership.** Run delivery for the mobile portfolio in Plane — maintain the backlog, coordinate developers, triage and reproduce bugs, chair reviews and set launch priorities and release dates across the archives, audio-visual, web and application teams.

Independent Consultant – Software & Data · Remote

May 2024 - Present

Client engagements delivered remotely alongside the roles above.

- **Web rebuild & reporting automation — Singapore.** Rebuilt and reworked pages across a client's three websites, and wrote a Python pipeline that generates formatted, print-ready PDF reports from their raw client records — replacing a manual assembly step.
- **Market research & outreach — Netherlands.** Researched and qualified target companies for a venture client, then ran the approved outreach and secured introductory meetings — a short, fully client-facing engagement delivered remotely.

Telecommunications Intern · Nepal Telecom, Kathmandu

Jun - Jul 2019

- Analysed and implemented a range of IP networking protocols, working through their concepts and practical behaviour across the state network.
- Monitored routes for data and voice transmission, and produced reports with senior engineers recommending network optimisation.

AI RESEARCH & PROJECTS

Explainable & Efficient Subgraph Retrieval in Graph RAG Systems · MSc thesis, LJMU

2025 - 2026

- **The problem.** Graph RAG grounds a model's answer in a subgraph pulled from a knowledge graph. Retrieve too narrowly and the reasoning path is missing; too broadly and structural noise drives the model into hallucination. The thesis targeted that trade-off.
- **What was built.** A layered pipeline combining *context-aware pruning* — discarding nodes and edges irrelevant to the question — with *relation-guided expansion*, growing the subgraph along the relation types the question implies rather than blindly by hop count, plus *dual-stage denoising* at retrieval and again before generation. Relation-guided two-hop retrieval proved the optimal scope — enough to preserve multi-step reasoning, tight enough to cut the noise that causes hallucination.
- **Explainability.** Built so each answer traces back to the specific subgraph that produced it — the property enterprise deployments require before they will trust generated output.

Neural Style Transfer · BE thesis, NSIT

2020

Deep convolutional network pipeline extracting and synthesising visual features to apply artistic styles to target photographs.

EDUCATION

MSc, Data Science	Liverpool John Moores University	Jan 2025 - Jan 2026	Merit (68%)
Executive PG Programme, Data Science	IIIT Bangalore	Jul 2023 - Aug 2024	3.84 / 4.0
BE, Electronics & Communication	Netaji Subhas Institute of Technology	Aug 2016 - Aug 2020	7.37 / 10

ADDITIONAL

COMPEX Scholarship (2016). Competitive full-tuition award, Embassy of India, Nepal.

Languages. English, Hindi, Nepali. · **Interests.** Long-term meditation practitioner; cricket, running, reading.